

Criteria	Extracted Value	Compliance Check	Status
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1. Grade of Concrete	M25 for all R.C.C works.	For Mangaluru, which falls under coastal regions of India, environmental exposure is typically 'Severe' as per IS 456:2000 Table 3 and Table 5. M25 concrete is generally acceptable for severe exposure conditions. The minimum grade of concrete for RCC in severe exposure conditions is M25.	Compliant

2. Reinforcement Bars	T/Y denotes HYSD TMT bars of Grade Fe 500 conforming to IS 1786-1985.	Fe 500 is a standard and widely used grade of reinforcement. No specific mention of shear reinforcement/stirrups grade, but it's generally implied to be the same unless specified.	Compliant
3. Lap Length	50 times dia of the bar, to be staggered such that not more than 50% of bars are lapped at a section.	A lap length of 50d is a standard design practice and is generally compliant with IS 456:2000 for tension laps where not less than 50% of bars are lapped at a section. SP 34 Appendix D (for development length, which influences lap length) gives minimum bond stress and development length formulas. For Fe 500 bars, 50d is a common and acceptable value assuming proper detailing.	Compliant
4. Clear Cover	Footing/Wall: 50 mm, Columns: 40 mm, Slab: 20 mm, Beam: 25 mm	Footing Cover: Minimum 50mm is explicitly mentioned and compliant. For footings, a 50mm cover means PCC is generally required beneath the footing to ensure the cover. The section "3 SECTION X" shows 125mm PCC below the footing. This is consistent. Column Cover: 40mm is compliant (within 40-70mm range). Beam Cover: 25mm is often used for beams, though IS 456 recommends 30mm for beams for moderate exposure condition. For severe exposure, 30mm or more would be preferred. Slab Cover: 20mm is typically used for slabs and is compliant.	Partially Compliant (Beam Cover)
5. Development Length (Ld)	50 times the dia. of the bar.	50 times the bar diameter (50d) is a common and generally compliant value for development length, especially for Fe 500. This is in line with standard practice derived from IS 456:2000.	Compliant

6. Safe Bearing Capacity (SBC) of soil	23.0 T/m ²	SBC value is provided. The drawing for Section X also shows 125mm PCC below the footing.	Compliant
7. Seismic Zone and Wind Load	Not explicitly mentioned in "NOTES" section. Project notes mention "DESIGNED FOR G+8 FLOORS".	This information is crucial for structural design, especially for a G+8 building. While the number of stories is mentioned, the specific seismic zone and wind load parameters are missing from the notes.	Missing Information
8. Building Limitations	Structure is designed for Ground+8 stories only.	The maximum number of stories is specified (G+8).	Compliant
9. Structure's Purpose	"CANTEEN EXTENSION BUILDING"	The purpose is clearly stated.	Compliant
10. Floor Heights	Ground Floor, First Floor (3.6M), Second to Eighth Floors (3.45M), Terrace.	Floor heights are indicated in the general layout.	Compliant
11. Schedule of Footings	Present, "SCHEDULE OF FOOTINGS" table.	The drawing includes a "SCHEDULE OF FOOTINGS" table with dimensions (Length, Width, Depth) and bottom reinforcement details for various footing marks (F1, F2, F2A, F3, F4, CF1, CF2, CF3). The layout shows these footing marks.	Compliant
12. Footing Type	Isolated footings (F1, F2, F2A, F3, F4, as seen in the layout and schedule). Combined Footings (CF1, CF2, CF3).	The drawing shows a combination of isolated and combined footings. For a G+8 structure, both types can be appropriate depending on column loads and soil conditions. The use of combined footings (CF1, CF2, CF3) suggests that adequate design considerations were made for closely spaced columns or boundary conditions.	Compliant

13. Reinforcement in High-Rise Buildings	Not explicitly mentioned in general notes or schedule of footings.	For a G+8 building, which is considered a medium-to-high rise structure, top reinforcement in footings is crucial, especially for eccentric loads, combined footings, or raft foundations. While combined footings (CF1, CF2, CF3) in the schedule show "TOP" reinforcement, isolated footings (F1, F2, F2A, F3, F4) only show "BOTTOM" reinforcement. This is a potential concern for isolated footings under non-uniform soil pressure or eccentric loading.	Partially Compliant (Isolated Footings)
14. Raft Foundation Reinforcement	Raft foundation is not used.	Not applicable as raft foundation is not the primary footing type.	N/A
15. Lift Design	No lift pit details are visible or mentioned.	Not specified in the provided drawings. Assuming no lift is designed based on the available information.	Not Specified/N/A
16. Soil Improvement	"IF LOOSE PATCHES FOUND SUITABLE THICKNESS OF BOULDER PACKING SHALL BE PROVIDE"	Soil improvement method (boulder packing) is mentioned for loose patches.	Compliant
17. Column Ties	"COLUMN TIES AS PER SCHEDULE" (in Section X detail). Column details show ties as closed loops.	The column schedules (C1-C8) show closed ties at specified spacing (e.g., Y8@200). In Section X, it shows "COLUMN TIES AS PER SCHEDULE". While continuous ties are visually implied by the closed loops, an explicit note about continuity would be ideal but not strictly mandatory if detailing is clear.	Compliant
18. Plan of Ties	No separate plan for ties is present. However, column schedules provide tie details.	While a separate plan would be good, providing detailed tie arrangements within the individual column schedules (C1-C8) is a common and acceptable practice.	Acceptable

19. Outer Ties Check	Column schedules C1-C8 show tie diameters and spacing (e.g., Y8@200). Percentage of steel in columns indicated visually by number and diameter of bars (e.g., 4Y16(a)+6Y12).	The percentage of steel is not explicitly stated in the notes but can be calculated from the column schedules. For columns, IS 456:2000 specifies a minimum of 0.8% and a maximum of 6% (or 4% if lapping). For C1-C8, typically the provided bars exceed 0.8%. Without exact dimensions for all columns, verification of the 0.8% minimum is not straightforward here but is assumed to be met if designed correctly.	Compliant (Assuming 0.8% min met)
20. Cross-Section Area	Not specified in "NOTES".	This is a general design principle but usually not explicitly stated in notes.	Missing Information
21. Steel Curtailment	Column schedules C1-C8 show variations in vertical bars from "FOUNDATION TO SECOND FLOOR" to "SIXTH FLOOR TO TERRACE LVL". For example, C1 changes from 4Y16(a)+6Y12 to 4Y16, and C3 changes from 6Y16 to 4Y16.	Curtailment of steel in upper floors is evident from the column schedules, typically reducing the number/diameter of bars as the load decreases. This is a standard and efficient practice.	Compliant
22. Maximum Steel Percentage in Columns	Not explicitly stated.	IS 456:2000 specifies a maximum of 6% (or 4% if lapping is present). This value isn't stated in the notes; calculation would be needed for each column based on its dimensions and reinforcement to confirm compliance. However, based on the number of bars shown (e.g. 4+6=10 bars in C1, sizes Y16, Y12), it's unlikely to exceed 4-6% for typical column sizes.	Missing Information (Explicit %)

Missing or Wrong Information

Clear Cover (Beam Cover): Beam clear cover is specified as 25mm. For severe environmental exposure (inferred for Mangaluru) as per IS 456:2000, 30mm or higher is generally recommended for beams.

Seismic Zone and Wind Load: Critical design parameters for a G+8 structure are missing from the notes.

Reinforcement in High-Rise Buildings (Isolated Footings Top Reinforcement): While combined footings show top reinforcement, isolated footings only specify bottom reinforcement in the schedule. For a G+8 building, explicit consideration or mention of potential top reinforcement for isolated footings under specific conditions would be prudent, as per best practices derived from IS 456:2000.

Cross-Section Area: Not explicitly stated whether gross or effective areas are used for design.

Maximum Steel Percentage in Columns: The maximum percentage of steel used in columns is not explicitly stated in the notes, which would ideally be a part of the design specifications.



Summary of Compliance:

The drawing is clearly identified as a foundation drawing and contains most of the necessary information in its "NOTES" section and various schedules. The site location can be inferred. The document adheres well to many aspects of IS 456:2000 and SP 34 regarding common design practices for concrete grade, reinforcement grade, lap length, clear cover for footings/columns, and development length.

However, the absence of explicit seismic zone and wind load data is a significant omission for a G+8 structure, as these heavily influence structural design. Additionally, the lack of specified top reinforcement for isolated footings in a multi-story building warrants attention.

Overall, the drawing appears to be a valid foundation drawing but has a few key pieces of information missing or needing clarification to ensure full compliance and best practice adherence.